N P Gayathri

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.20
Kendriya Vidhyalaya Yelahank Air force Station	Class XII (CBSE)	2019	78.00%
Kendriya Vidhyalaya DRDO	Class X (CBSE)	2017	10.00%

TECHNICAL SKILLS

- **Geotechnical Laboratory Techniques:** Unconfined Compressive Strength (UCS) testing, sieve analysis, Atterberg limits, standard Proctor compaction testing, specific gravity determination, soil classification (USCS), and direct shear testing.
- **Soil Suction & Hydraulic Characterization:** Handling of HYPROP and dew point potentiometer (WP4C) instruments for soil water retention and suction measurements, and soil-water characteristic curve (SWCC) analysis.
- **Design & Drafting:** AutoCAD (2D/3D drafting, building drawings), STAAD.Pro (structural analysis and RCC/steel design).
- **Structural & Materials Testing:** Concrete and highway materials testing, strength-of-materials testing, and surveying (total station and theodolite).
- **Data Analysis:** MS Excel for experimental data processing, calculations, and graphical representation.

EXPERIENCE

Intern -- *Larsen & Toubro (L&T) Construction, Prestige Finsbury Park* | Jul 2022 - Aug 2022

- Completed a site internship covering execution and monitoring of building construction activities, including preparation of snag lists and observation of ongoing site works.
- Rated sincere and hardworking by site HR management during the internship tenure.

PROJECTS

Experimental Investigation of Biopolymer-Stabilized Soil: Effects on Soil-Water Characteristic Curves and Water Retention Behavior *(M.Tech Thesis)* | 2026 - Present

Department of Civil Engineering, IIT Hyderabad

- Conducting experimental investigations on biopolymer-treated soils to quantify changes in soil-water characteristic curves (SWCC) and water retention behavior.
- Evaluating the influence of biopolymer type and dosage on soil hydraulic properties.
- Assessing the feasibility of biopolymer stabilization as a sustainable alternative to conventional soil improvement techniques.

Contaminated Soil Stabilisation Using Lime *(B.Tech Final Year Project)* | 2022 - 2023

School of Civil Engineering, REVA University -- Guide: Dr. Pradeep Kumar B K

- Investigated stabilization/solidification (S/S) of municipal-solid-waste-contaminated soil collected from a dump-yard site using lime as a stabilizing additive.
- Characterized soil through sieve analysis, Atterberg limits, specific gravity, and compaction tests, classifying the soil as well-graded clayey sand (SW-SC).
- Evaluated stabilized soil performance at 4%, 6%, and 8% lime dosages across 0, 5, 7, and 14-day curing periods using unconfined compressive strength (UCS) testing.
- Concluded that lime stabilization did not meet IRC 2015 strength criteria for pavement/construction use due to weak soil-stabilizer bonding.

CERTIFICATIONS

- Construction Project Management -- Columbia University

- Construction Scheduling -- Columbia University
- Managing Project Risks and Changes -- University of California, Irvine
- Budgeting and Scheduling Projects -- University of California, Irvine

POSITION OF RESPONSIBILITY

Teaching Assistant -- *IIT Hyderabad*

- Served as a Teaching Assistant for third-year B.Tech students in Highway Engineering and Materials, supporting course delivery and student learning.
- Served as a Teaching Assistant for the Geotechnical Engineering Laboratory course for second-year B.Tech students, assisting with lab sessions and student learning.

Postgraduate Representative -- *Civil Engineering Society, IIT Hyderabad* | Present

- Represent postgraduate students in the Civil Engineering Society, facilitating communication between students and the society to support academic, technical, and professional engagement.
- Contribute to the coordination of society activities and initiatives, encouraging postgraduate participation and collaboration within the Civil Engineering community.

PG Buddy -- *Sunshine, IIT Hyderabad* | 2026

- Contributed to the planning and execution of student well-being and engagement initiatives, including the Sunshine Happiness Carnival 2026, Random Acts of Kindness, Painting Without Brushes, and Art Therapy activities.

EXTRACURRICULARS AND ADDITIONAL SKILLS

- Interested in interior design, sketching, and creative visualization tools
- Playing Badminton, Cycling, Doodling

SCHOLASTIC ACHIEVEMENTS

- Graduated B.Tech in Civil Engineering with First Class with Distinction (CGPA: 9.28/10), REVA University
- Achieved a perfect CGPA of 10.0 in Class X (CBSE), Kendriya Vidyalaya
- Consistently secured "O" and "A+" grades across multiple semesters of B.Tech coursework